

Lesson 37: Chain, Product & Quotient Rules — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Differentiate $y = (x + 1)^3$.
2. Differentiate $y = (2x - 1)^4$.
3. Differentiate $y = (5x + 2)^2$.
4. Which rule for $(3x+1)^6$?
5. State the product rule.
6. Which rule for $(x+1)/(x-1)$?

Section 2 — Core practice

7. Differentiate $y = (3x + 2)^4$.
8. Differentiate $y = (x^2 + 1)^3$.
9. Differentiate $y = x^2(x + 1)$ (product rule).
10. Differentiate $y = (4x - 3)^5$.
11. Differentiate $y = x(2x + 1)$ (product rule).

12. Differentiate $y = (1 - x)^4$.

Section 3 — Challenge & exam-style

13. Differentiate $y = (2x^3 + 1)^4$.

14. Differentiate $y = x^2(x - 1)^3$ (product + chain).

15. Differentiate $y = x/(x + 1)$ (quotient rule).

16. Differentiate $y = (3x - 1)/(x + 2)$ (quotient rule).